

Merchant Revenue Forecasting

Selecting a standard method for 12-month merchant revenue planning

Self-directed portfolio project using synthetic data

BUSINESS PROBLEM
Which method should Finance use as the standard 12-month merchant revenue forecast, and when should an individual merchant use a different method?

PORTFOLIO 50 synthetic merchants	HISTORY 60 months (2020–2024)	OBSERVATIONS 3,000 merchant-months	PRIMARY METRIC Unweighted mean MAPE
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Finance needs a repeatable forecasting process across merchants with different growth, seasonality, and volatility. The 2024 evaluation uses 48 months of training history and a 12-month test period. Backtesting also evaluates forecasts for 2022 and 2023 using information available at each forecast date. Seasonal Naïve is the baseline. MAPE is an unweighted average across merchants rather than weighted by revenue. Driver Scenario Regression uses planning assumptions available at the forecast date rather than actual results from the test period.

2024 HOLDOUT RESULT
Holt produced the lowest average merchant-level MAPE in the 2024 test period at **8.64%**, compared with **12.47%** for Seasonal Naïve. That is a 3.83 percentage-point improvement, or a 30.7% reduction in forecast error.

30.7%
relative reduction versus Seasonal Naïve on the 2024 holdout

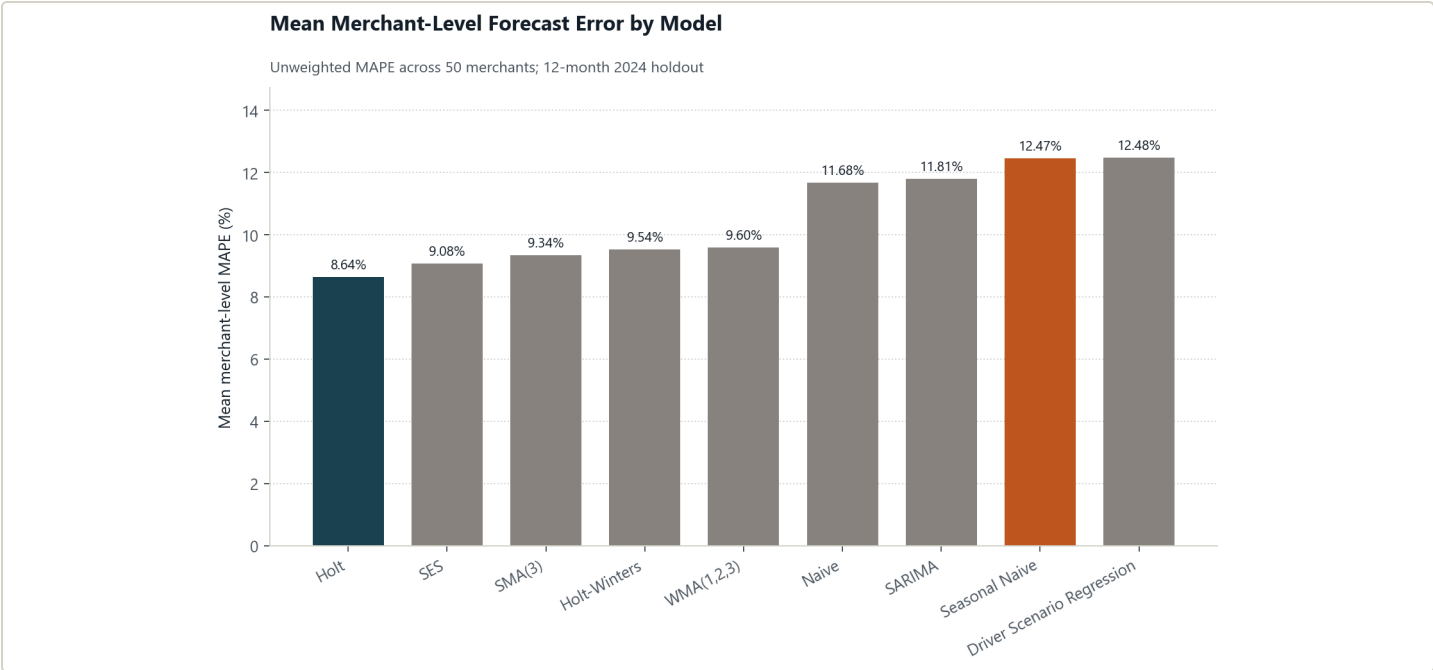


Figure 1. Average merchant-level MAPE across 50 merchants for the 12-month 2024 test period. Holt leads the 2024 comparison; Seasonal Naïve is the baseline benchmark.

Management interpretation: More complex methods were not necessarily more accurate. Driver Scenario Regression produced 12.48% average MAPE and SARIMA produced 11.81%, while simpler smoothing methods performed better in 2024. Holt was the best method for 17 of 50 merchants (34%). However, backtesting across 2022–2024 shows that SES was slightly more consistent overall, so Finance should reassess the standard method during each forecast cycle.

Recommendation and Merchant-Level Exceptions

Use Holt for the current plan, compare alternatives each cycle, and manage merchant-level exceptions

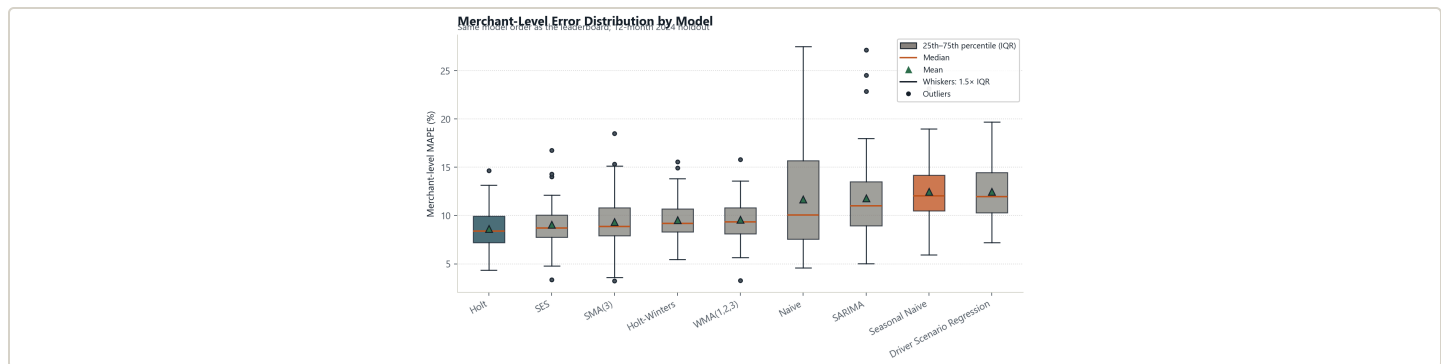


Figure 2. Merchant-level MAPE distribution in the same model order as Figure 1. Holt combines low average error with a relatively contained distribution, although forecast accuracy still varies by merchant.

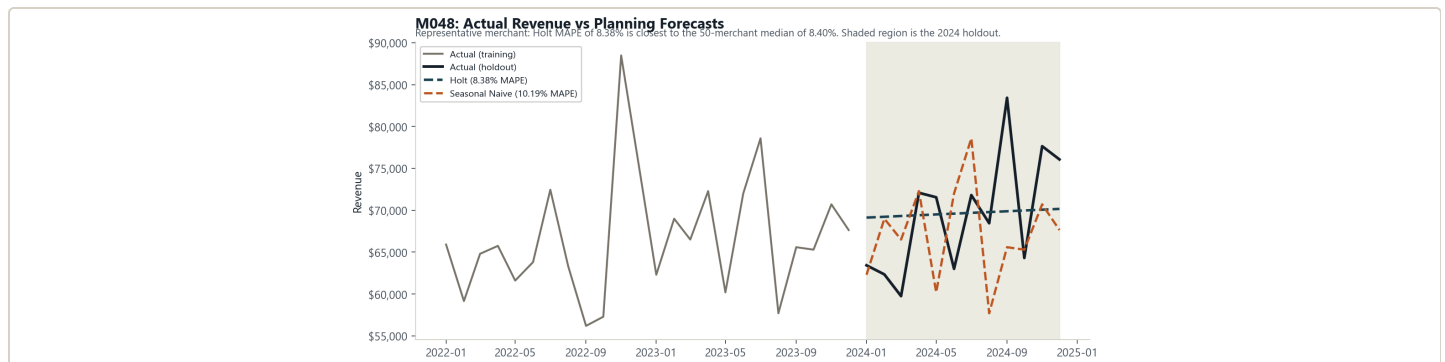


Figure 3. Merchant M048 is shown because its Holt MAPE of 8.38% is closest to the 50-merchant median of 8.40%. The chart compares Holt with the Seasonal Naive baseline for the 2024 test period.

Recommended forecasting process

- Use Holt as the standard method for the current planning cycle, but reassess it each cycle. SES had the lower average error across the 2022–2024 backtests: 9.75% versus 9.88% for Holt, primarily because Holt weakened to 12.31% in the 2022 forecast period.
- Retain Seasonal Naive as the baseline benchmark during every forecast cycle.
- Compare Holt, SES, and SMA(3) using the same forecast date and 12-month horizon.
- Use a merchant-specific method only when it produces consistently lower error across multiple forecast periods.
- Place merchants with persistently high or volatile forecast error on an exception list and use wider forecast ranges, business-partner input, and explicit upside and downside scenarios.

Limitations

- The data is synthetic and does not represent an employer or customer portfolio.
- Results may not generalize to actual merchant businesses.
- MAPE is averaged equally across merchants and is not weighted by portfolio revenue.
- Driver-based forecasts depend on planning assumptions available at the forecast date.
- Model rankings can change as new actuals become available; reassess performance through periodic backtesting.
- SARIMA and Driver Scenario Regression were not included in the 2022 comparison because they require 36 months of training history.
- This case study does not claim production deployment or realized business savings.

Management takeaway: Finance should favor the most repeatable and consistently accurate forecasting process, not the most complex model. A standard method, a clear baseline, periodic backtesting, and disciplined merchant-level exceptions provide stronger planning control than selecting a method based on complexity alone.